LING406_Lec13_Aphasias_Notes

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Section 1: Reasoning About Aphasia Before Chomsky

- **Aphasia** is a loss or disruption of language ability caused by brain damage. In the **1870s**, a group of German neurologists, including **Lichtheim** and **Wernicke**, began making more **systematic observations** of aphasic patients. Instead of only recording that a patient had "lost language," they tried to explain each specific pattern of loss using a model of how language is organized in the brain.
- The core idea of these neurologists was that the brain contains separate **"centers"** for different language functions, and that these centers are linked to one another by pathways. A specific loss of function could then be explained as a **"disconnection"** between two centers. In a disconnection, each center might still work on its own, but information can no longer travel from one to the other, so any task that depends on that route breaks down.
- **Lichtheim's "House" model** (described in Eling & Whitaker, 2009, *History of aphasia: from brain to language*, in the *Handbook of Clinical Neurology*, Vol. 95, pp. 571–582) is the classic version of this approach. It gets its name from its shape, which looks like a house with a pointed roof. The model contains three centers and two outside pathways.
    - **A** is the **center for sound images**, which stores the sound patterns of words. When we hear a word, its sound pattern is recognized here.
    - **M** is the **center for movement images**, which stores the movement patterns needed to say words out loud.
    - **B** is the **center for concepts**, sitting at the top of the "roof." It holds meanings, and it connects to both A and M.
    - **a** is the **acoustic pathway**, which carries incoming sound up into A. **m** is the **motor pathway**, which carries commands down from M to produce speech.
- The model connects these parts with lines: A connects to M along the floor of the house, A connects up to B, and M connects up to B. The numbers **1 through 7** mark places where damage would produce a different **form of aphasia**, depending on whether a center or a pathway is destroyed.
    - Damage at **1** destroys the movement center M itself, and damage at **2** destroys the sound center A itself.
    - Damage at **3** cuts the direct line between A and M. Damage at **4** cuts the line between M and B, and damage at **6** cuts the line between A and B.
    - Damage at **5** cuts the motor pathway leaving M, and damage at **7** cuts the acoustic pathway entering A.
- The logic of the model lets us predict a specific symptom from a specific break. A **loss of the ability to understand spoken words** might be due to a **disconnection between A and B**. In that case, the sound of a word still arrives through the acoustic pathway and is still recognized as a sound pattern in A, but that pattern can no longer reach the concept center, so the word is heard without being understood.
- In the same way, a **loss of the ability to do verbatim repetition** may be due to a **disconnection between A and M**. Repeating a word exactly requires the sound pattern recognized in A to be passed directly to the movement patterns in M, which then send it out through the motor pathway. If that direct line is cut, a patient could still understand

words and still speak from their own ideas, since the routes through B remain intact, but could fail at simply saying back what they just heard.

- This kind of **"box and arrow" model** is foundational, not only in **aphasiology** (the study of aphasia) but also in modeling **typical language function** in healthy people. The **interactive activation model** is another example of this style of modeling, in which language is described as a set of processing units connected by pathways that pass information between them.

Section 2: The Geschwind-Wernicke Model

- In the **1970s**, **Norman Geschwind** revived this centers-and-connections approach. The resulting account, known as the **"Geschwind-Wernicke model" of brain and language**, places the language centers on real anatomical regions of the left side of the brain and links them with a real bundle of nerve fibers.
    - **Broca's area** sits in the lower part of the frontal lobe, toward the front of the brain.
    - **Wernicke's area** sits farther back, in the temporal lobe.
    - The **arcuate fasciculus** is a curved bundle of fibers that arches from Wernicke's area forward to Broca's area, connecting the two.
    - Two major grooves serve as landmarks. The **central sulcus**, also called the **fissure of Rolando**, runs down from the top of the brain and separates the frontal lobe from the region behind it. The **lateral sulcus**, also called the **fissure of Sylvius**, runs along the side of the brain and separates the temporal lobe below it. A **sulcus** is a groove in the brain's surface, and **fissure** is an older name for a large groove.
- In the century between Wernicke and Geschwind, the field of **linguistics evolved dramatically**. The most important change was the development of a theoretical model in which **syntax**, the system of rules for how words combine into sentences, was **independent of other aspects of language**. In this model, syntax had its own **computational principles**, meaning its own rules for building structure that did not depend on what the words meant or how they sounded.
- This new view of syntax **opened the door for theorizing about the (double) dissociations of language functions** in more sophisticated ways. A **dissociation** occurs when brain damage impairs one function while leaving another intact. A **double dissociation** occurs when one patient has function X impaired and function Y spared, while another patient has the reverse pattern. Double dissociations are strong evidence that two functions rely on separate systems, and once linguists described syntax as its own system, researchers could ask whether syntax specifically could be lost on its own.

Section 3: Fromkin's Description of Broca's and Wernicke's Aphasia

- **Victoria A. Fromkin**, in her 1997 paper *Some thoughts about the brain/mind/language interface* (Lingua, Vol. 100, pp. 3–27), gives a standard description of the two best-known aphasia types.
- The speech output of **Broca's aphasia** patients is characterized by three features. There are **word-finding pauses**, where the speaker stops while searching for a word. There is a **loss of grammatical morphemes**, which are the small words and word parts that carry grammatical information, such as "the," "is," or the endings "-ed" and "-s." Quite often there is also **disturbed word order**, so words come out in an arrangement that does not follow the rules of the language.
- **Auditory comprehension** in Broca's aphasia, meaning the understanding of heard speech, gives the impression of being generally good in **colloquial conversation**, which is ordinary everyday talk. However, **controlled testing reveals considerable impairments**. Casual conversation can hide comprehension problems that become visible only when a test is carefully designed to measure understanding.

- The term **agrammatism**, which refers to speech that lacks normal grammatical structure, is **almost synonymous with Broca's aphasia**. The overlap is not perfect in either direction, though. Some patients with lesions in Broca's area are not agrammatic, and some agrammatic patients would not be classified neurologically as Broca's aphasics.
- **Wernicke's aphasia** patients, on the other hand, produce **fluent speech** with **good intonation**, meaning the natural rise and fall of the voice, and good **pronunciation**. Their speech contains many **word substitutions**, where one word is replaced by another, and these substitutions can be either **semantically similar** (close in meaning to the intended word) or **semantically dissimilar** (unrelated in meaning). They also produce **neologisms**, which are made-up words that do not exist in the language, as well as **phonological errors**, which are errors in the sounds of words. Unlike the description of Broca's patients in casual conversation, Wernicke's patients also show clear **comprehension difficulties**.

Section 4: An Attractive Dichotomy: Word Salad Versus Telegraphic Speech

- These descriptions suggest an **attractive dichotomy**, a clean two-way split between the two aphasia types. **Wernicke's aphasia** is described as **"word salad"** or **"Jabberwocky,"** often with **neologisms**, while **syntax remains roughly intact**. "Word salad" captures the way words are tossed together without a coherent meaning, and "Jabberwocky" refers to speech that has the sound and shape of real sentences but does not make sense.
- One Wernicke's patient, asked about his health, replied: "I felt worse because I can no longer keep in mind from the mind of the minds to keep me from mind and up to the ear which can be to find among ourselves" (**Kriendler et al., 1971**).
    - The reply is built from pieces that are grammatical on their own, such as "I felt worse," "keep in mind," and "up to the ear," and it flows like one long sentence.
    - Despite this grammatical surface, the pieces do not combine into any coherent meaning about his health, which is exactly the pattern of intact syntax with disrupted word selection.
- Another patient, asked about his poor vision, said: "My wires don't hire right." The sentence has a perfectly normal grammatical frame, with a subject, a negated verb, and an adverb at the end. The words placed into that frame are wrong, and "hire" rhymes with "wires," so the error has a sound-based relationship to the surrounding words rather than a meaningful one.
- An **aphasic physician**, asked if he was a doctor, replied: "Me? Yes, sir, I'm, a male demaploze on my own. I still know my tubaboys what for I have that's gone hell and some of them go."
    - "Demaploze" and "tubaboys" are **neologisms**, invented words that fill noun positions in otherwise normal phrase frames such as "a male ___" and "I still know my ___."
    - The speaker produces fluent, grammatical-sounding phrases while the meaning breaks down, which matches the description of Wernicke's aphasia as fluent but uninterpretable.
- One aphasic patient described a fork as "a need for a schedule" and a spoon as "How many schemes on your throat." These descriptions are fluent strings of real words, but they do not describe the objects. The word "throat" in the spoon description has a loose connection to eating, which fits the description of substitutions that range from semantically related to unrelated.
- **Broca's aphasia**, on the other side of the dichotomy, is described as **"telegraphic" speech**, with **difficulty with word order and grammatical particles** and a **relative sparing of important content words**. **Grammatical particles** are the small function words, such as articles, prepositions, and helping verbs. **Content words** are the words that carry the main meaning, such as nouns and main verbs. The term "telegraphic" comes

from old telegrams, where people paid by the word and so left out everything except the essential words.

- A Broca's aphasic, asked what brought him to the hospital, answered: "Yes – ah – Monday – ah – Dad and P.H. (the patient's name) and Dad ... hospital. Two ... ah doctors ..., and ah ... thirty minutes ... and yes ... ah ... hospital. And, er Wednesday ... nine o'clock and eh Thursday, ten o'clock ... doctors. Two doctors ... and ah ... teeth. Yeah ... fine" (**Badecker and Caramazza, 1985**).
    - The content words survive: "Monday," "Dad," "hospital," "two doctors," "thirty minutes," "Wednesday," "nine o'clock," "Thursday," "ten o'clock," and "teeth." From these alone, a listener can piece together a rough story about going to the hospital with his father, seeing doctors, and having something done involving teeth.
    - Almost all of the grammatical glue is missing. There are almost no verbs, no tense markers, no articles, and no prepositions linking the words, and the speech is broken up by long pauses and filler sounds such as "ah" and "er."
    - These halting but partly interpretable utterances are very different from the fluent but uninterpretable utterances of Wernicke's aphasia, and this contrast is what makes the dichotomy so attractive.

Section 5: Lexicon Versus Grammar, and Broca's Original Patient

- The dichotomy can be stated in more theoretical terms. **Wernicke's aphasia** is described as **problems with activation (or perhaps inhibition) in the lexicon**, and **Broca's aphasia** is described as **problems with grammar**.
    - The **lexicon** is the mental store of words. A problem with **activation** means the wrong words become active and get selected, which fits the word substitutions and neologisms of Wernicke's aphasia.
    - The question mark next to **inhibition** reflects an alternative possibility: the problem may be a failure to suppress competing words, so that many words stay active and the wrong one wins.
    - A problem with **grammar** means difficulty with the rules that combine words into structured sentences, which fits the missing grammatical morphemes and disturbed word order of Broca's aphasia.
- This grammar-based description is **fairly distant from how Broca described his original cases**. Broca's most famous patient was **Leborgne**, who was nicknamed **"Tan."** Broca's **behavioral description** of Tan had three parts.
    - Tan **could only produce one jargon syllable**, the syllable "tan" that gave him his nickname.
    - He **seemed otherwise to be in possession of his senses**, meaning his other mental and sensory abilities appeared normal.
    - He **seemed to understand what was being said** to him.
- Broca's presentation of Tan raises three considerations about the reasoning behind the case.
    - The first concerns the **role of dissociations** in the argument. It is critical that Leborgne had his other senses intact, since only then can the loss of speech be separated from a general loss of mental function. If he had also been unable to hear, think, or understand, his silence could have been one part of an overall decline rather than evidence for a separate speech ability.
    - The second asks why a **"capacity for articulate speech"** sounds like a more plausible function of a particular brain region than **"criminality"** or **"philoprogenitiveness,"** which means love of one's offspring. Articulate speech is a specific, observable ability that can be lost while other abilities remain, as Tan's case demonstrates, whereas criminality and love of offspring are broad

social and personality traits that are much harder to define as a single function with its own location.
    - The third asks what measurements we would want to make if we could examine Tan, in order to specify his deficits in terms of the **levels of description used in modern linguistics**. These levels include the **phonetic** level (the physical production of speech sounds), the **phonological** level (the sound system and sound patterns of the language), the **lexical** level (words), and the **syntactic** level (sentence structure).
- The mismatch here is genuinely confusing: **Broca's actual patients would not necessarily be classified as Broca's aphasics** today. They were arguably **unable to produce speech at all**, apparently due to **motor control issues**, rather than producing speech with missing grammar. Today they would be classified as suffering from **apraxia of speech**, a disorder in planning and coordinating the movements needed to produce speech, even though the muscles themselves can move.

Section 6: Grammar and Lexicon in Formal Linguistics

- The Wernicke's-lexicon and Broca's-grammar dichotomy is interesting in part because it seems **well aligned with a sharp distinction drawn in late 20th century formal linguistics** between the **grammar** and the **lexicon**. If the brain respects this same division, then damage to one region should impair the grammar while damage to another should impair the lexicon.
- This distinction starts with **Chomsky's** formulation of the **computational problem of language learning**, along with his compelling demonstrations of the **independence of syntax and semantics**. **Semantics** is the meaning of words and sentences, so the independence claim is that the rules of sentence structure can operate separately from meaning.
- Chomsky's view holds that we recognize a new string of words as a sentence not because it matches some familiar item in any simple way, but because it is **generated by the grammar** that each individual has somehow internalized. We understand a new sentence, in part, because we are capable of determining the **process by which that sentence is derived** in this grammar.
    - The important consequence is that a speaker's knowledge of language cannot simply be a memorized list of sentences heard before, since people constantly recognize and understand sentences they have never encountered.
    - Instead, each speaker has an **internalized grammar**, a set of rules that can produce an unlimited number of sentences, and understanding a sentence involves working out how those rules built it.
- Chomsky's famous example (**Chomsky, 1955**) contrasts two word strings: **(a) "colorless green ideas sleep furiously"** and **(b) "furiously sleep ideas green colorless."** Neither (a) nor (b), nor any of their parts, has ever occurred in the past linguistic experience of an English speaker, yet (a) is grammatical while (b) is not.
    - Sentence (a) is meaningless: ideas cannot be green, something cannot be both colorless and green, and ideas do not sleep. Even so, it follows a normal English pattern of adjective, adjective, noun, verb, adverb, and English speakers judge it to be a proper sentence.
    - Sentence (b) uses exactly the same words in reverse order, and speakers immediately judge it to be ungrammatical.
    - The two strings are equally meaningless and equally unfamiliar, so the difference in judgment must come from syntax alone. This is the evidence that syntax operates independently of semantics.

Section 7: Functional and Positional Levels in Speech Production

- The same grammar-lexicon split appears in the difference between the **Functional** and **Positional levels** of the speech production model. This model describes speaking as a series of processing levels, each building on the one above it.
    - The **Message Level** is **conceptual**, not yet linguistic. Here the speaker chooses the **story angle**, arouses **concepts of the referents** (the things being talked about), and sets up the **event structure**, meaning who is doing what in the situation.
    - The **Functional Level** is the **first linguistic level**. Here the speaker arouses **word meanings** and assigns **semantic roles**, such as which participant is the doer of the action and which is acted upon.
    - The **Positional Level** is the **second linguistic level**. Here the speaker gets **word forms into grammatically appropriate order**, building the actual sentence structure.
    - The **Phonological Encoding Level** fills in the sounds of the words.
    - The **Speech Gesture Level** is **articulation**, the physical movements of the mouth, tongue, and vocal cords that produce speech.
- **Lemmas**, the word entries selected at the functional level, have an **arbitrary relationship to their referents**. Nothing about the word "dog" resembles an actual dog, so the connection between the word and its meaning must simply be learned and stored. This makes lemmas the lexicon side of the dichotomy.
- In contrast, the **syntactic trees**, **affixes**, and other structures at the positional level are the product of **abstract syntactic processes**. An **affix** is a word part attached to a stem, such as "-ly" or "-ed." These structures are built by rules rather than stored one by one, which makes the positional level the grammar side of the dichotomy.
- A natural objection concerns words like **HOUSEBOAT** and **DOGHOUSE**, where the meaning is clearly **not arbitrary**. A houseboat is a boat that serves as a house, and a doghouse is a house for a dog, so the meanings follow from the parts. The answer comes from **speech error evidence** that **derivational morphology** is done at the **positional level**.
    - **Derivational morphology** is the process of building new words from parts, such as adding "-ly" to an adjective to form an adverb.
    - The speech error **"easy enoughly"** comes from an intended "easily enough." The "-ly" affix, which belonged on "easy," ended up attached to "enough" instead.
    - For an affix to land on the wrong word, it must be handled as a separate piece and placed into position at the stage where words are being ordered. This places word-building at the positional level, so words whose meanings come from their parts are assembled by grammatical processes rather than stored as arbitrary lemmas.

Section 8: Sentence Processing and Parsing

- There are also popular theories of **sentence processing** in which the **syntactic structure** of sentences is **computed independently of meaning** as we listen or read. On these theories, the comprehension system first builds a grammatical structure for the incoming words, and meaning is fitted to that structure.
- These structure-building mechanisms are called **"parsing" mechanisms**, and **isolating them requires carefully constructing stimulus materials**. Ordinary sentences give many clues to their meaning besides syntax, so researchers must design sentences in which syntax alone determines the interpretation.
- The sentence **"The burglar blew open the safe with the dynamite"** has two possible structures, each represented as a **syntactic tree**. In these trees, **S** stands for sentence, **NP** for noun phrase, **VP** for verb phrase, **det** for determiner (a word like "the"), **N** for noun, **V** for verb, **PP** for prepositional phrase, and **prep** for preposition.

    - In the **preferred** structure **(A)**, S splits into an NP, "The burglar," and a VP. The VP contains three parts side by side: the verb "blew open," the NP "the safe," and the PP "with the dynamite." The prepositional phrase is attached to the **verb phrase**, so it describes an aspect of the action, namely the **instrument**: the burglar used the dynamite to blow the safe open.
    - In the **unpreferred** structure **(B)**, the VP contains the verb "blew open" and one larger NP. That larger NP contains "the safe" together with the PP "with the dynamite." The prepositional phrase is now attached to the **noun phrase**, so it describes a property of the safe, namely what was in it: the burglar blew open the safe that contained dynamite.
    - The words and their order are identical in both versions. The only difference is where the prepositional phrase attaches in the tree, and that single structural difference produces two different meanings.
- Under these theories, **one possibility** is that **Broca's patients generally understand what's being said** even though they have a **core grammatical deficit**. Their comprehension in conversation could look good if they rely on cues other than syntax, while the grammatical system itself is damaged.

Section 9: Broca in the 1860s and the Problem of Testing Syntax

- None of this theory was available to Broca back in the **1860s**. For Broca, it was interesting enough to **dissociate "articulate speech" from comprehension**, which is why the observation that Tan **seemed to understand what was being said** carried so much weight in his argument.
- Demonstrating that some patients have a difficulty that is **more generally with syntax** is a harder problem. It requires the kind of measurement Broca could not make: specifying a patient's deficit at the **syntactic level** in particular, separate from the phonetic, phonological, and lexical levels. A patient who speaks with great effort or barely speaks at all cannot reveal much about syntax through production, so the evidence has to come from a task that measures grammar in some other way.

Section 10: Why Comprehension Deficits in Broca's Aphasia Matter

- Showing that Broca's patients have **comprehension deficits** was important for a specific reason. **Broca's aphasia is almost always accompanied by some level of motor dysfunction**. If **agrammatic speech merely reflects an attempt to communicate despite motor limitations**, then the missing grammar in their speech tells us about their muscles and movement planning, not about their knowledge of grammar.
- **Hughlings Jackson** made this point long ago: when we study a patient, we are observing **what the individual can do despite the impairment**. Just because a brain injury leads to agrammatic speech does not literally mean that **the grammar itself is injured**. The speech we hear is the output of a damaged system working as well as it can, and the patient may be producing the best message possible under difficult conditions.
- **Chomsky's distinction between competence and performance** applies directly here. **Competence** is a speaker's underlying knowledge of the language, including the grammar. **Performance** is the actual use of that knowledge in speaking and understanding, which can be disrupted by things like fatigue, memory limits, or motor problems. When motor deficits obviously interfere with performance in speaking, **comprehension may provide a clearer window into competence**, since understanding a sentence does not require producing any movements.
- **Caramazza & Zurif (1976)** describe the view that dominated before their work. Researchers of the time (citing Kellar, 1973, and **Lenneberg, 1973**) accepted that **anterior damage**, meaning damage to the front of the brain where Broca's area lies, **spares the patient's tacit knowledge of his language**. **Tacit knowledge** is knowledge a person has

and uses without being consciously aware of it, such as knowing the rules of grammar without being able to state them.

- One version of this view, from **Lenneberg (1973)**, attributes the **telegrammatic output** to a **neuromuscular problem**.
- On this account, the **strain of speaking is so great** that the patient speaks **asyntactically**, leaving out grammatical structure, **in order to economize effort**. The patient keeps only the most informative words because every word costs so much effort to produce.
- If this account were correct, Broca's patients should have normal comprehension of grammar, and finding a grammatical comprehension deficit would challenge it.

Section 11: Man Bites Dog: Why Special Tests Are Needed

- **Psycholinguistic tests** are needed to observe grammatical difficulties in comprehension, since in ordinary conversation there are **lots of redundant cues** to what a speaker intends. A listener can often arrive at the right meaning without using syntax at all, so normal-looking comprehension in conversation does not prove that the grammar is intact.
- Consider the sentence **"The dog bit the man."** There are many ways to get the right meaning here.
  - The first is to take the **most likely combination** of DOG, BITE, and MAN and form an **event representation**, either **with or without syntax**. Dogs biting people is a familiar event and people biting dogs is not, so world knowledge alone points to the correct interpretation.
  - The second is **canonical word order**. In English, the usual order is **Agent – Verb – Undergoer**, which in grammatical terms is **Subject – Verb – Object**. The **agent** is the one who performs the action, and the **undergoer** is the one the action happens to. A simple strategy of treating the first noun as the doer and the noun after the verb as the receiver gives the right answer for this sentence.
- In addition, a lot of language comprehension takes place in **discourse context**, the surrounding conversation and situation, which helps a lot. Knowing what has already been discussed narrows down what a new sentence is likely to mean.
- Broca's patients' comprehension might depend on these **compensatory strategies** rather than on grammar. If that were the case, we would want to test them with sentences more like **"The dog that the man is biting is angry."**
  - In this sentence the **man** is the one doing the biting, and the **dog** is the one being bitten and the one who is angry.
  - World knowledge pushes toward the wrong answer, since a dog biting a man is far more likely than a man biting a dog.
  - The word-order strategy also fails, since "the dog" is the first noun and would be taken as the doer.
  - Only correct **syntactic analysis** of the embedded clause "that the man is biting" yields the right interpretation, so a patient who relies on likely events or canonical word order will make systematic errors on sentences of this kind. This reversal of the expected event, where the man bites the dog, is the source of the label **"Man Bites Dog."**

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Section 12: Caramazza and Zurif's Sentence Materials

- **Caramazza and Zurif** built a list of exactly this kind of sentence, along with a set of **control sentences**, in a study titled **"Dissociation of Algorithmic and Heuristic Processes."** An **algorithmic process** in comprehension is full, rule-based syntactic

analysis that works out the structure of a sentence step by step. A **heuristic process** is a shortcut, such as guessing the meaning from which event is most likely or from the usual word order. The study was designed to separate these two kinds of processing by building sentences where the shortcuts either help, give no information, or lead to the wrong answer.

- The sentence materials fall into four groups of eight sentences each, for a total of 32 sentences.
    - **A. Control sentences (1–8)** are simple sentences with no embedding, such as "The girl is kicking a green ball," "The cat is chasing a blue bird," and "The man is reading a yellow book." The doer comes first and the object comes after the verb, so both syntax and the shortcuts lead to the correct meaning.
    - **B. Center-embedded sentences with semantic constraints (9–16)** contain a clause placed in the middle of the main sentence, such as "The apple that the boy is eating is red," "The wagon that the horse is pulling is green," and "The balloon that the clown is holding is round." These sentences are syntactically complex, but meaning constrains the answer: a boy can eat an apple, while an apple cannot eat a boy. A listener can therefore get the right interpretation from word meanings alone, without analyzing the structure.
    - **C. Reversible center-embedded sentences (17–24)** have the same structure but use two nouns that could each perform the action, such as "The cow that the monkey is scaring is yellow," "The cat that the dog is biting is black," and "The girl that the boy is tickling is happy." A monkey can scare a cow and a cow can scare a monkey, so meaning gives no help, and only syntax tells the listener who did what.
    - **D. Improbable center-embedded sentences (25–32)** describe events where the syntax produces an unlikely meaning, such as "The horse that the girl is kicking is brown," "The bird that the worm is eating is blue," and "The dog that the man is biting is black." Here common sense pushes toward the wrong interpretation, since birds eat worms far more often than worms eat birds, so the listener must follow the syntax against expectation.
- In every center-embedded sentence, the first noun is the one acted upon, while the second noun is the doer. In "The cow that the monkey is scaring is yellow," the monkey does the scaring and the cow is scared and yellow. A word-order shortcut that treats the first noun as the doer therefore fails on all of the center-embedded sentences, and the three center-embedded groups differ only in whether meaning helps, gives no clue, or misleads.

Section 13: Results of the Sentence-Picture Matching Task

- The data come from a **sentence-picture matching task**, in which a participant hears a sentence and picks the picture that matches it. Performance is measured as the **mean number of correct responses** for each sentence type, out of a possible 8, since each type contains eight sentences. A flat reference line sits at 4, which is half of the 8 items. Four groups were tested: **N (Normal)**, **B (Broca)**, **C (Conduction)**, and **W (Wernicke)**, across four sentence types: **C (control)**, **CE$_S$ (semantically constrained center-embedded)**, **CE$_R$ (reversible center-embedded)**, and **CE$_I$ (improbable center-embedded)**.
- **Healthy controls** are more or less **equally good at all stimulus types**, scoring about 8 out of 8 on controls, semantically constrained, reversible, and improbable sentences alike. For a healthy listener, the syntax of a center-embedded sentence is easy to process, so whether meaning helps or misleads makes little difference.
- **Wernicke's patients are poor overall**, scoring below the other patient groups on the simple control sentences at about 6, and dropping to about 5 on the improbable sentences, their lowest point. This overall weakness is probably a result of their general **comprehension deficits**, which affect even simple sentences rather than only syntactically complex ones.

- The **Broca's patients** score close to normal on control sentences, at nearly 8, and stay high on the semantically constrained sentences, at a little over 7.5. Their scores then fall to about 6.3 on reversible sentences and about 6.5 on improbable sentences. The conduction group follows a similar shape, staying around 7.4 on control and semantically constrained sentences and dropping to about 6.2 on both reversible and improbable sentences.
    - On the semantically constrained sentences, Broca's patients can use word meanings to reach the right answer, so their syntactic weakness stays hidden.
    - On reversible and improbable sentences, meaning either gives no help or points the wrong way, so the listener must rely on syntax, and this is exactly where Broca's patients fail.
- The Broca's patients therefore have a particular difficulty **keeping "who did what to whom" straight** when **syntax conflicts with common sense**, with the **probable semantic roles** for the nouns, and with similar cues. A **semantic role** is the part a noun plays in an event, such as doer or receiver of the action, and "probable" semantic roles are the ones a listener would expect from world knowledge.
- This **specific comprehension deficit** was taken as evidence that **Broca's aphasia impacts syntax specifically**. The deficit appears only when syntax is the sole route to the answer, which fits a damaged grammatical system rather than a general comprehension problem or a purely motor problem with speaking. It also challenges Lenneberg's neuromuscular account, since a speaking problem should not produce errors in understanding heard sentences.

Section 14: Grammar Versus Lexicon as a Structure-Function Question

- A more specific theory of what the **"parts of language"** might be opens up new ways of exploring **structure-function relationships**, meaning the relationships between particular brain structures and the particular language functions they support. A theory that names syntax as its own system gives researchers a precise target to test in patients.
- Rather than just dividing aphasic patients into those with **good and poor comprehension**, or those who are **more or less fluent**, researchers can isolate problems with **particular aspects of language**. The main division of this kind is **the grammar versus the lexicon**, with Broca's aphasia linked to the grammar and Wernicke's aphasia linked to the lexicon.

Section 15: Problems With the Dichotomy: The Dick et al. Task

- **Dick et al.** tested this dichotomy by having participants listen to **four different types of sentences**. In their task (Experiments 1A and 1B), participants faced a computer screen with two pictures side by side, such as a cat and a dog, and responded with a button box that had a button on each side. The **instructions** were to **indicate which animal is doing the action**.
- The four sentence types all describe the same event, in which the dog bites the cat.
    - An **active** sentence: *"The dog is biting the cat."* The doer comes first, in the canonical Subject – Verb – Object order.
    - A **subject cleft**: *"It's the dog that is biting the cat."* A **cleft sentence** splits a simple sentence into two parts using "It's ... that" to highlight one noun. In a subject cleft, the highlighted noun is the subject, so the doer still comes first and the canonical order is kept.
    - An **object cleft**: *"It's the cat that the dog is biting."* The highlighted noun is the object, so the first noun heard, the cat, is the one being bitten. The order of agent and undergoer is reversed from the canonical pattern.
    - A **passive**: *"The cat is bitten by the dog."* The undergoer comes first as the grammatical subject, and the doer appears later after "by."
- A listener using the shortcut of treating the first noun as the doer would answer correctly on actives and subject clefts and incorrectly on object clefts and passives. Every sentence

is also **reversible**, since a dog can bite a cat and a cat can bite a dog, so world knowledge gives no help. Only syntax reliably identifies the doer in the object cleft and passive sentences.

Section 16: Aphasic Patients on the Dick et al. Task

- **Aphasia patients** and **healthy aging controls** did this task, with patients **grouped by the Western Aphasia Battery**, a standard assessment used to sort patients into aphasia types. Performance is measured as the **percentage of correct responses** for each sentence type, and the error bars represent **±1 SEM**, the **standard error of the mean**, which indicates how precisely each group average is estimated. With two pictures to choose from, a participant guessing at random would score around 50%.
- The results by group follow clear patterns.
  - The **elderly controls** score near ceiling throughout, at about 98% on actives, about 100% on subject clefts, about 97% on object clefts, and about 95% on passives.
  - The **anomic** patients score about 90% on actives and 88% on subject clefts, then drop to about 69% on object clefts and recover partly to about 73% on passives.
  - The **conduction** patients score about 75% on actives and 73% on subject clefts, then fall to about 43% on object clefts and about 50% on passives.
  - The **Broca's** patients score about 61% on actives and 65% on subject clefts, then fall to about 42% on object clefts and rise to about 53% on passives.
  - The **Wernicke's** patients stay low across every type, at roughly 44% on actives, 46% on subject clefts, 43% on object clefts, and 50% on passives.
- A pattern emerges that is much like what **Caramazza and Zurif** observed. The **Broca's aphasia patients** have **particular difficulty with the syntactically complex stimuli**, the object clefts and passives, where their scores drop to around or below the guessing level. However, their **performance is overall not so great**, since even on the simple actives they reach only about 61%, far below the elderly controls. Their difficulty is therefore not limited to complex syntax, which weakens the claim that their deficit is purely grammatical.

Section 17: Healthy Listeners Under Stress

- Dick et al. also tested **healthy undergraduates** under conditions that **stress the language system in different ways**. These **"stress" conditions** included things like **speeding up the speech** and asking people to **remember lists of numbers as a secondary task** while they listened.
- In the **single-stress conditions** (Experiment 1B), each participant faced one kind of stress at a time.
  - **Normal** listening produced scores near 99% to 100% on every sentence type, dipping only to about 98% on object clefts.
  - **Auditory Digits** and **Visual Digits**, in which participants held a list of numbers in memory that they either heard or saw, lowered object cleft scores only slightly, to around 97.5%, and passive scores to about 97%.
  - The **Low-Pass Filter** condition, which removes the higher frequencies of the speech and makes it sound muffled, lowered object clefts to about 96% and passives to about 97%.
  - The **Compressed** condition, in which the speech was sped up, kept actives and subject clefts near 99% but dropped object clefts to about 92% and passives to about 95%.
  - The **Noise** condition, in which background noise was added, produced the largest single-stress effect, with actives at about 97%, subject clefts at about

94%, object clefts at about 86%, and passives at about 92%.

- In the **dual-stress conditions**, two kinds of stress were combined.
  - **Compressed + Visual Digits** produced about 97.5% on actives and 99.5% on subject clefts, with object clefts falling to about 90.5% and passives to about 96%.
  - **Noise + Visual Digits** produced about 98% on actives and 97% on subject clefts, with object clefts at about 89% and passives at about 95%.
  - **Low Pass + Compression** produced by far the largest drop. Actives fell to about 91% and subject clefts to about 89.5%, while object clefts fell to about 65.5% and passives to about 68%.
- Across every manipulation, the **object cleft and passive sentences are particularly vulnerable**. Under the hardest condition, low-pass filtering combined with compression, healthy young adults fall roughly 25 percentage points on object clefts and passives compared with actives, which is the same shape of deficit seen in the Broca's patients: relatively good on canonical sentences and much worse on noncanonical ones.
- These healthy participants have intact grammars, so their errors cannot come from a loss of grammatical knowledge. The errors come from limits on processing, such as degraded input and a divided memory load. This suggests that the **comprehension deficits seen in Broca's aphasia patients** may be a **"performance" issue after all**, in Chomsky's sense, rather than damage to grammatical competence. Sentences that depart from canonical word order are simply harder to process, and any strain on the language system, whether from brain damage or from difficult listening conditions, exposes that difficulty first.

Section 18: Are Broca's Aphasics' Comprehension Deficits Language-Specific?

- **Saygin et al. (2004)** asked whether the comprehension problems of Broca's aphasics are limited to language. Patients were asked to **match an action to an object**. In the non-linguistic version, a picture of a boy pantomiming an action appears alongside two object pictures, and the correct match is an ice cream cone.
  - The distractor objects differ in how they relate to the target. A **cake** is related in meaning, being another sweet food, which makes it a **semantic** distractor.
  - A **bouquet of flowers** is held up to the face in a similar way, which makes it an **affordance-based** distractor. An **affordance** is the way an object is handled or used.
  - A **rooster** has no meaningful relationship to the action, which makes it an **unrelated** distractor.
- Accuracy is reported for the non-linguistic version and for a linguistic version of the same task.
  - On the **non-linguistic** version, controls scored **98.5%** and aphasic patients scored **93.4%**.
  - On the **linguistic** version, controls scored **99.4%** and aphasic patients scored **87.5%**.
- Accuracy also varied by distractor type.
  - Controls scored **99.8%** with unrelated distractors, **99.2%** with semantic distractors, and **97.7%** with affordance-based distractors, so they were near ceiling in every case.
  - Aphasic patients scored **93.3%** with unrelated distractors, **86.3%** with semantic distractors, and **91.6%** with affordance-based distractors. Their largest drop came with semantic distractors, where the wrong object was closest in meaning to the right one.
- **Broca's aphasics had difficulty with this task whether the stimuli were linguistic or not.** Their deficit was larger in the linguistic version, but they were still below controls when the task involved only pictures of a gesture and objects, with no words at all. A

purely grammatical deficit should leave a wordless pantomime task untouched, so this result suggests that the problems extend beyond language.

Section 19: Lesion Correlates of Pantomime Interpretation Deficits

- Saygin et al. also related performance on the **non-linguistic (pantomime interpretation)** version to the location of each patient's brain damage. The lesion maps are three horizontal slices through the brain, numbered **1** (lowest) to **3** (highest), with the left hemisphere colored. **CS** marks the **central sulcus**, the groove separating the frontal lobe from the regions behind it.
- The strongest relationship, marked in red, appears in slice 2, in the **frontal region in front of the central sulcus**. Weaker relationships, in green and blue, spread across surrounding parts of the left hemisphere.
- **Performance in the non-verbal version** of this task was **significantly correlated with damage to Broca's Area**. Patients with more damage in Broca's area made more errors in interpreting pantomimed actions, even though no language was involved. This links Broca's area to a function that is not specific to language.

Section 20: Are Wernicke's Aphasics' Comprehension Deficits Language-Specific?

- Saygin et al. also asked patients to **match environmental sounds to pictures**. In the **non-verbal** version, a participant hears a sound such as a cow mooing and chooses between pictures such as a cow and a sheep. In the **verbal** version, the participant hears a spoken phrase such as "cow mooing" and chooses between pictures such as a cow and a violin.
- Accuracy for each group in the verbal and nonverbal versions was as follows.
  - **Controls**: **99.2%** verbal and **98.1%** nonverbal.
  - **RHD**, which stands for **right hemisphere damage**: **99.1%** verbal and **98.2%** nonverbal.
  - **Anomic** patients: **98.6%** verbal and **96.7%** nonverbal.
  - **Broca's** patients: **96.3%** verbal and **95.2%** nonverbal.
  - **Wernicke's** patients: **90.0%** verbal and **91.3%** nonverbal.
- The groups differed in their overall accuracy at $P < 0.0001$, which means differences this large would be extremely unlikely if the groups truly performed the same. The ordering of the groups was:

$$\text{Control} = \text{RHD} = \text{Anomic} > \text{Broca's} > \text{Wernicke's}$$

- All of these comparisons held at a **corrected** $P < 0.01$, where "corrected" means the significance threshold was adjusted to account for making many comparisons at once.
- **Wernicke's patients were particularly bad at this task**, with the lowest scores of any group. Just as important, **no significant difference was observed between the verbal and non-verbal versions** of the task: their score on nonverbal sounds, 91.3%, was essentially the same as their score on spoken phrases, 90.0%. A deficit confined to the lexicon should impair spoken phrases but spare the recognition of a cow's moo, so this result suggests that the comprehension problem in Wernicke's aphasia extends to meaningful sounds in general.

Section 21: Lesion Overlays for Sound-Picture Matching

- **Lesion overlays based on accuracy** combine the brain damage of all patients who were impaired on a given version of the task, so the warmest colors, red, mark the regions damaged in the most impaired patients. There are two sets of overlays, each on three horizontal slices: one for patients **impaired in NONVERBAL** sound matching ($n = 8$) and one for patients **impaired in VERBAL** sound matching ($n = 10$).

- In both sets, the area of greatest overlap sits in the **back part of the left temporal region**, with red concentrated toward the rear of the hemisphere and yellow and teal spreading outward from it. The two sets of maps look strikingly alike.
- **Deficits in both verbal and non-verbal sound-picture matching are associated with Wernicke's area**. The same region is linked to failures in understanding spoken words and failures in recognizing environmental sounds, which again points to a deficit that is not specific to language.

Section 22: How Language-Specific Are the Deficits in Aphasia?

- How language-specific the deficits in aphasia really are is **still a controversy**. Evidence from Caramazza and Zurif supports a specifically grammatical deficit in Broca's aphasia, while the Dick et al. stress studies and the Saygin et al. non-linguistic tasks suggest that the deficits reflect processing limits and impairments that reach beyond language.
- The terms of the debate have changed greatly from the 19th century. The debate now rests on three kinds of evidence.
    - **Relatively sophisticated tests** designed to tap **specific aspects of language**, such as reversible center-embedded sentences and object clefts that isolate syntax from meaning and word order.
    - **Quantitative analyses of behavior**, such as accuracy percentages compared across groups and sentence types with error bars and significance tests.
    - **Quantitative analyses of structure-function relationships**, such as statistical correlations between task performance and the location of brain damage.
- This is a long way from 19th century methods, when Broca's case of Tan rested on a behavioral description of a single patient and a dissociation between articulate speech and comprehension.

Section 23: Voxel-Based Lesion-Symptom Mapping

- **Voxel-based lesion-symptom mapping** (Elizabeth **Bates**, Stephen M. **Wilson**, Ayse Pinar **Saygin**, Frederic **Dick**, Martin I. **Sereno**, Robert T. **Knight**, and Nina F. **Dronkers**) is a method for **formalizing structure-function relationships in aphasia data** using **MRI data**. A **voxel** is a tiny three-dimensional unit of a brain image, like a pixel with depth. At each voxel, the method asks how strongly damage at that spot is related to a symptom, and the strength of that relationship is expressed as a ***t* value**, with higher values meaning a stronger association.
- The **"symptoms"** in this analysis are defined **very broadly**. Rather than testing specific aspects of grammar, the analysis looks at overall **fluency** and overall **comprehension**.
    - Maps **(a–c)** mark regions likely to be damaged in individuals with **fluency deficits**. The *t* values range from about 1.7, in blue, to 8.5, in red, and the strongest values fall in the **anterior**, or front, part of the left hemisphere.
    - Maps **(d–f)** mark regions associated with **comprehension deficits**. The *t* values range from about 1.8, in blue, to 7.6, in red, and the strongest values, clearest in the lowest slice (d), fall in the **posterior**, or back, part of the left temporal region.
    - The slices come from three heights in the brain: (a) and (d) are the lowest, (b) and (e) are in the middle, and (c) and (f) are the highest.
- Data from a **large number of patients** are consistent with a **rough double dissociation**. **Posterior regions** are associated with **perception**, since damage there predicts comprehension deficits, whereas **anterior regions** are associated with **production**, since damage there predicts fluency deficits. The dissociation is called "rough" because the symptoms are defined so broadly and the colored regions spread and overlap, yet the overall split between front and back matches the classic division between Broca's and Wernicke's aphasia.